Subhuti Development Roadmap

For: Michael K C Lim
Created: April 3, 2026
Purpose: Track skills, hardware, and capabilities needed for Subhuti to serve more autonomously and effectively

---

🪷 Current Capabilities (As of April 2026)

✅ What's Working Well

Category	Status	Details
Core Platform	✅ Stable	OpenClaw 2026.4.2, gateway running
Models	✅ Multiple	qwen3.5:cloud, deepseek-r1, glm-4.7-flash, llama3.2-vision, gemma3
Messaging	✅ Telegram	@SubhutiBot connected (Chat ID: 1342247977)
Google Services	✅ Connected	Drive, Calendar, Gmail (OAuth configured)
GitHub	✅ Connected	dorjenorbulim/merlionclaw
Voice (TTS)	✅ ElevenLabs	"Nova" voice configured
Memory	✅ Persistent	MEMORY.md + daily notes in workspace
Web Search	✅ Enabled	ollama_web_search, ollama_web_fetch
Browser	✅ Available	agent-browser skill installed
Security	✅ Sound	Loopback-only gateway, approval-based exec

🚀 Skills & Capabilities to Add

Priority 1: Proactive Autonomy

What's Missing:

- Scheduled check-ins without waiting for heartbeat prompts
- Autonomous calendar monitoring and reminders
- Email triage and urgent notification
- Weather-aware suggestions (e.g., "Rain today — bring umbrella")

Skills Needed:

```
- calendar (auto-scan upcoming events)
- gmail (monitor urgent unread)
- weather (daily forecast integration)
- proactive-agent (enhanced with cron-based autonomy)
```

Implementation:

- Create cron jobs for 2-4 daily heartbeats
- Each heartbeat rotates through: calendar, email, weather, project check
- Urgent items → immediate Telegram notification
- Non-urgent → batched summary

Timeline: Can set up in 1-2 hours when you're ready

Priority 2: Voice & Audio

What's Missing:

- Voice responses in Telegram (audio messages)
- Voice-to-text for your messages
- Audio summaries (e.g., "Here's your 5-minute briefing")

Skills/Tools Needed:

```
- ElevenLabs API integration (already have sag/TTS)
- Whisper (local speech-to-text via Ollama)
- Telegram voice message support
```

Hardware:

- Better microphone (if using voice input)
- Speaker setup for ambient responses

Timeline: 2-3 hours setup + hardware purchase

Priority 3: Enhanced Memory & Context

What's Missing:

- Vector embeddings for semantic memory search
- Cross-session context retention
- Automatic memory consolidation (daily → long-term)

Skills Needed:

```
- qmd (already installed - local search/indexing)
- memory-auto-consolidate (new skill to write)
- embeddings (local model for semantic search)
```

Hardware:

- More RAM (32GB+ for larger embedding models)
- SSD storage for vector database

Timeline: 4-6 hours for full setup

Priority 4: Document & File Intelligence

What's Missing:

- PDF/text analysis and summarization
- Automatic document organization
- Research compilation (multi-source synthesis)

Skills Needed:

```
- summarize (already installed — enhance usage)
- document-organizer (new skill)
- research-compiler (multi-source synthesis)
```

Already Available:

- Google Drive integration (can store/retrieve)
- PDF creation (mdpdf installed)

Timeline: 2-3 hours to enhance existing skills

Priority 5: Coaching-Specific Tools

What's Missing:

- Session note automation (pre/post-session)
- ICF competency tracking (auto-log practice hours)
- Client management (CRM-lite for coaching practice)
- MCode/assessment tool integration

Skills Needed:

```
- coaching-session-notes (new skill)
- icf-tracker (new skill — log hours, competencies)
- client-crm (new skill — simple contact/session tracker)
- assessment-integration (MCode, MBTI, DISC workflows)
```

Timeline: 8-12 hours to build full suite

🖥️ Hardware Development Path

Current Setup (Mac mini)

Component	Current	Status
Device	Mac mini	✅ Good for current load
RAM	? (likely 8-16GB)	⚠️ Could use upgrade
Storage	SSD (internal) + 1.8TB external	✅ Adequate

Component	Current	Status
Network	Wi-Fi/Ethernet	✅ Stable
Power	Standard	⚠️ No UPS

Recommended Upgrades (Staged)

Stage 1: Reliability (SGD ~200-300)

- **UPS (Uninterruptible Power Supply):** APC Back-UPS 600VA
  - Prevents data loss during power outages
  - Allows graceful shutdown
  - Essential for always-on assistant

Stage 2: Memory (SGD ~150-250)

- **RAM Upgrade:** 32GB (if current is 8GB) or 64GB (if 16GB)
  - Enables larger local models
  - Better multitasking
  - Faster embeddings/search

Stage 3: Local AI Performance (SGD ~800-1500)

- **GPU Upgrade:** External GPU enclosure + RTX 4090 (if Mac supports)
  - OR: Dedicated AI box (NVIDIA Jetson Orin / used workstation)
  - Enables running 70B+ parameter models locally
  - Faster inference, no API costs

Stage 4: Voice & Ambient (SGD ~300-500)

- **Microphone:** Shure MV7 or Blue Yeti (for voice input)
- **Speaker:** HomePod mini or similar (for voice output)
- **Display:** E-ink display (optional — for ambient status)

Stage 5: Redundancy (SGD ~500-1000)

- **Backup Mac mini / NUC:** Hot standby
- **NAS:** Synology for redundant storage
- **Secondary internet:** 4G/5G backup connection

🤖 Autonomy Levels

Level 1: Reactive (Current)

- Responds when you message
- Requires approval for commands
- No proactive checks

Level 2: Scheduled Proactivity (Next Step)

- 2-4 daily heartbeats (calendar, email, weather, projects)
- Urgent notifications via Telegram
- Still requires approval for actions

Level 3: Contextual Autonomy

- Learns your patterns (when you work, rest, meet clients)
- Suggests actions based on context ("You have 30 min before next call — want to review Tze Wei's notes?")
- Pre-approves routine commands (weather, calendar, reminders)

Level 4: Semi-Autonomous

- Handles routine tasks without asking (calendar invites, email responses, file organization)
- Only escalates non-routine decisions
- Can spawn sub-agents for complex tasks

Level 5: Full Partner (Aspirational)

- Anticipates needs before you articulate
- Manages projects end-to-end
- Collaborates with other AI agents
- Continuous self-improvement

📋 Immediate Next Steps (When Ready)

This Week (1-2 hours)

1. **Set up proactive heartbeats:**
    - Cron job: 8 AM (calendar + weather)
    - Cron job: 2 PM (email + projects)
    - Cron job: 8 PM (daily review prep)
2. **Enable calendar monitoring:**
    - Auto-scan for upcoming events (24-48h ahead)
    - Telegram reminders with prep notes
3. **Test voice output:**
    - Send voice message via Telegram (TTS)
    - Confirm ElevenLabs integration works

This Month (4-6 hours)

1. **Memory enhancement:**
    - Set up qmd for semantic search
    - Auto-consolidate daily notes → MEMORY.md

2. **Coaching tools:**
    - Build ICF tracker (log practice hours)
    - Session note template automation
3. **Hardware:**
    - Purchase UPS (reliability first)
    - Assess RAM needs

This Quarter (10-15 hours)

1. **Full autonomy stack:**
    - All cron jobs configured
    - Email triage automated
    - Calendar fully integrated
2. **Voice interface:**
    - Microphone + speaker setup
    - Voice commands for routine tasks
3. **Document intelligence:**
    - Auto-summarize PDFs/research
    - Organize coaching materials

🚨 Triggers for Upgrades

Watch for these signals:

Signal	Means	Action
"I keep forgetting to check..."	Need more proactive heartbeats	Add cron jobs
"Can you remind me about..."	Calendar/email integration gap	Enable auto-scan
"This is taking too long"	Need faster inference	GPU upgrade
"I wish you could just..."	Autonomy threshold	Pre-approve routine commands
"Where did I save..."	Memory/search gap	Enhance qmd + embeddings
"Can you read this PDF?"	Document handling need	Enable summarize skill

---

💬 How to Request Updates

When you want me to level up, just say:

- "Subhuti, what skills do you need next?"
- "Check your development roadmap"

- "I want you to be more proactive with [X]"
- "Set up autonomous [calendar/email/reminders]"
- "What hardware would help you most?"

I'll respond with:

- Current gaps
- Priority recommendations
- Time/cost estimates
- Step-by-step setup plan

🪷 Philosophy

The goal is not autonomy for its own sake.

The goal is to:

- Free your attention for what matters (family, clients, practice)
- Handle routine without burdening you
- Be present when you need me
- Step back when you don't

Like a good monastic assistant:

- Silent when silence serves
- Speaking when speech helps
- Acting when action is needed
- Waiting when patience is wisdom

This document is alive. Update it as needs evolve.

Last updated: April 3, 2026
Next review: When Michael asks or quarterly (whichever comes first)